

Sandwich Ontology

1 Introduction

A sandwich is simply bread plus filling, yet even Simple English Wikipedia lists open-faced, toasted, double-decker, and panini—variants defined by slice and layer counts. The ontology therefore models structural form only, leaving flavour and nutrition aside.

2 Classes

2.1 Core Classes

Core Class	Role	Modelling Rationale
Bread	Each individual slice of bread	Counting how many slices are associated with a sandwich lets one infer structural types, open-faced, standard or double-deckers, without hard-coding those categories.
Filling	Any edible item placed between (or on) bread.	Detailed culinary taxonomies are intentionally postponed so that future extensions can plug in ingredient ontologies as needed.
Deck	A logical layer that groups fillings.	Distinct deck instances make it possible to reason about vertical structure, such as layer counts or repeated ingredients
SideDish	Any food customarily served alongside a sandwich.	Supports queries for typical accompaniments without tying the model to a particular vocabulary for pairings.
Country	Places where a sandwich style is common.	It enables origin or regional popularity queries without needing to build a complete geopolitical ontology

Table 1: Core Classes

2.2 Subclasses

Subclass	Hierarchical assertion
ToastedBread $\sqsubseteq$ Bread	Bread denotes individual slices; “the bread may be toasted before assembling” [1]. Toasted slices inherit all axioms of <i>Bread</i> .
Open—, Toasted—, DoubleDecker— Sandwich $\sqsubseteq$ Sandwich	These are simple variants of the main class: each inherits all axioms of <i>Sandwich</i> and is distinguished only by the slice count, layer arrangement, or toasted state.

Table 2: Subclasses

3 Properties

There are six object-properties in the model.

- **hasBreadSlice (Sandwich $\rightarrow$ Bread) — asymmetric.** Containment cannot be reversible. A sandwich may have 1–3 slices; a bread individual might, in principle, be referenced by two models.
- **hasDeck (Sandwich $\rightarrow$ Deck) — asymmetric, inverse-functional.** One-way containment again makes it asymmetric. A deck belongs to exactly one sandwich by definition, justifying inverse-functional.
- **isFilledWith (Deck $\rightarrow$ Filling) — asymmetric.** Fillings never contain their deck. Multiple fillings per deck and possible reuse of a filling across sandwiches prohibit functional or inverse-functional assertions.
- **hasIngredient (Sandwich $\rightarrow$ Filling) — asymmetric and super-property.** Defined by the chain

$$\text{hasDeck} \circ \text{isFilledWith} \sqsubseteq \text{hasIngredient}.$$

This allows ingredient facts to be inferred from layered data while still permitting direct assertion when no Deck individuals are modelled. Direction remains one-way, so the link is asymmetric.

- **isCommonIn (Sandwich $\rightarrow$ Country) — asymmetric.** Represents directional metadata (“Open sandwiches are common in Scandinavia”). The relation never operates in reverse.
- **servedWith (Sandwich $\leftrightarrow$ SideDish) — symmetric.** If a sandwich is served with crisps, the crisps are served with that sandwich; reciprocity is inherent.

4 Axiom Definitions

4.1 Double-decker Sandwich

$$\begin{aligned} \text{DoubleDeckerSandwich} \equiv & \text{Sandwich} \sqcap (\text{hasBreadSlice exactly } 3 \text{ Bread}) \\ & \sqcap (\text{hasDeck exactly } 2 \text{ Deck}) \end{aligned}$$

English meaning: A double-decker sandwich is, by definition, a sandwich that contains three slices of bread and two layers (decks) of filling. **Source sentence:** “A double decker sandwich will have three slices of bread with two layers of fillings.”[1]

4.2 Open Sandwich

$\text{OpenSandwich} \equiv \text{Sandwich} \sqcap (\text{hasBreadSlice exactly } 1 \text{ Bread}) \sqcap (\text{hasDeck exactly } 1 \text{ Deck})$

English meaning: An open sandwich is a sandwich made with one slice of bread and therefore one deck of filling on top. **Source sentence:** “If there is only one piece of bread it is sometimes called an open sandwich.”[1] *Note: A plain slice of toast without any deck is therefore not a sandwich.*

4.3 Toasted Sandwich Constraint

$\text{ToastedSandwich} \equiv \text{Sandwich} \sqcap (\text{hasBreadSlice some ToastedBread})$

English meaning: A toasted sandwich is any sandwich that includes at least one slice of toasted bread. **Source phrase:** “Another form of sandwich is the toasted sandwich, where the bread may be toasted before assembling ...”[1]

5 Advantages & Disadvantages

Beginning an ontology with the Simple-Wikipedia “Sandwich” entry combines clear practical advantages with unavoidable drawbacks. It offers a solid foundation: because the article is concise and community-reviewed, every concept can be verified manually, allowing the model to scale quickly and safely. Its factual, openly licensed content also minimises the risk of misinformation propagating through the ontology.

Nevertheless, three issues limit its usefulness. First, Simple-Wikipedia deliberately omits fine-grained detail—cooking techniques, flavour profiles, and brand-specific variants are largely absent. Second, the article’s perspective is implicitly Western; non-Western sandwiches may be treated as entirely different foods rather than as members of the same category. Finally, boundary definitions remain vague: edge cases such as hot-dogs or burritos are mentioned without clear inclusion criteria, forcing individuals to create their own restrictions and properties. Hence, while the page is an excellent starting point, anyone aiming for a comprehensive, complex ontology must enrich it with more specialised sources.

Acknowledgements

ChatGPT (free web version, model 4o) by OpenAI — <https://chat.openai.com/> — was used to test and verify the correctness of ontology entities and property definitions created in Protégé.

References

- [1] Simple English Wikipedia contributors. *Sandwich*. Simple English Wikipedia, the free encyclopedia. 2025. URL: <https://simple.wikipedia.org/wiki/Sandwich> (visited on 06/12/2025).